

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 50.680340212265314, median 34.792694916979926, std: 44.60661354295471
Gyroscope error (imu0): mean 26.200554817060652, median 16.571284619496886, std: 38.9125848897061
Accelerometer error (imu0): mean 15.096542928551951, median 11.78973151700496, std: 13.292553450913658

Residuals

Reprojection error (cam0) [px]: mean 101.36068042453063, median 69.58538983395985, std: 89.21322708590942
Gyroscope error (imu0) [rad/s]: mean 0.21218479265219653, median 0.1342022951620437, std: 0.31513297386385536
Accelerometer error (imu0) [m/s^2]: mean 1.8547380106624187, median 1.448468254194621, std: 1.6331026421647301

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.98309817 0.04485647 -0.17749896 -0.00229371]
[0.16848708 -0.15762578 -0.97301912 -0.0409774]
[-0.07162461 -0.9864796 0.14740388 0.10451697]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.98309817 0.16848708 -0.07162461 0.01213521]
[0.04485647 -0.15762578 -0.9864796 0.09674765]
[-0.17749896 -0.97301912 0.14740388 -0.05568513]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)

0.006656712091050809

Gravity vector in target coords: [m/s^2]

[1.34949747 -9.49254115 2.05886912]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [588.5774585397844, 586.4301014972124]

Principal point: [327.0189377139115, 242.2584623130127]

Distortion model: radtan

Distortion coefficients: [0.11962886627460176, -0.2674498243556465, -0.0009995862977424511, -0.0012116942957377276]

Type: aprilgrid

Tags:

Rows: 6

Cols: 9

Size: 0.029 [m]

Spacing 0.0087 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 95

Accelerometer:

Noise density: 0.012605012

Noise density (discrete): 0.12285845967784913

Random walk: 0.0018792068913671173

Gyroscope:

Noise density: 0.000830887

Noise density (discrete): 0.008098484712775285

Random walk: 0.0001619178948911119

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

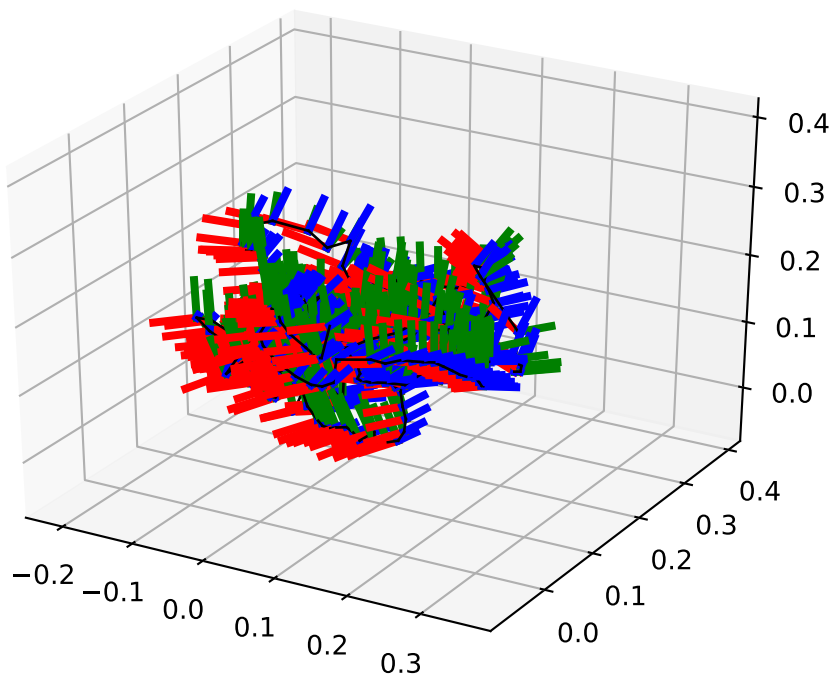
[0. 1. 0. 0.]

[0. 0. 1. 0.]

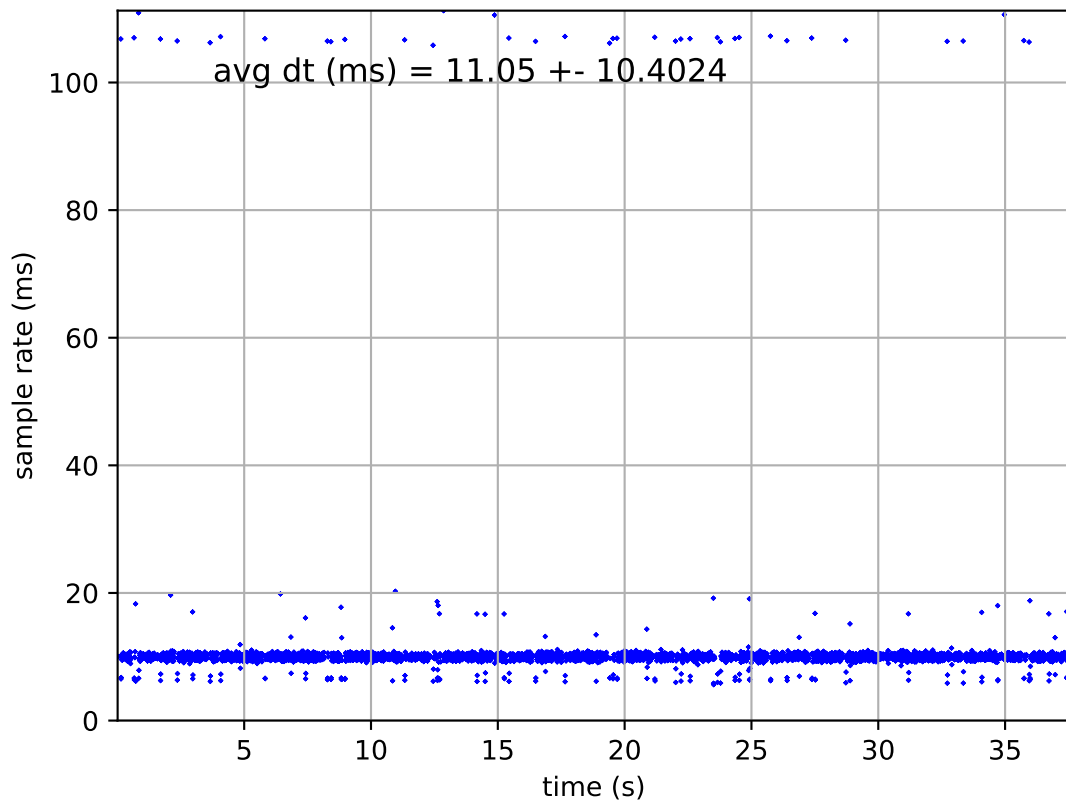
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

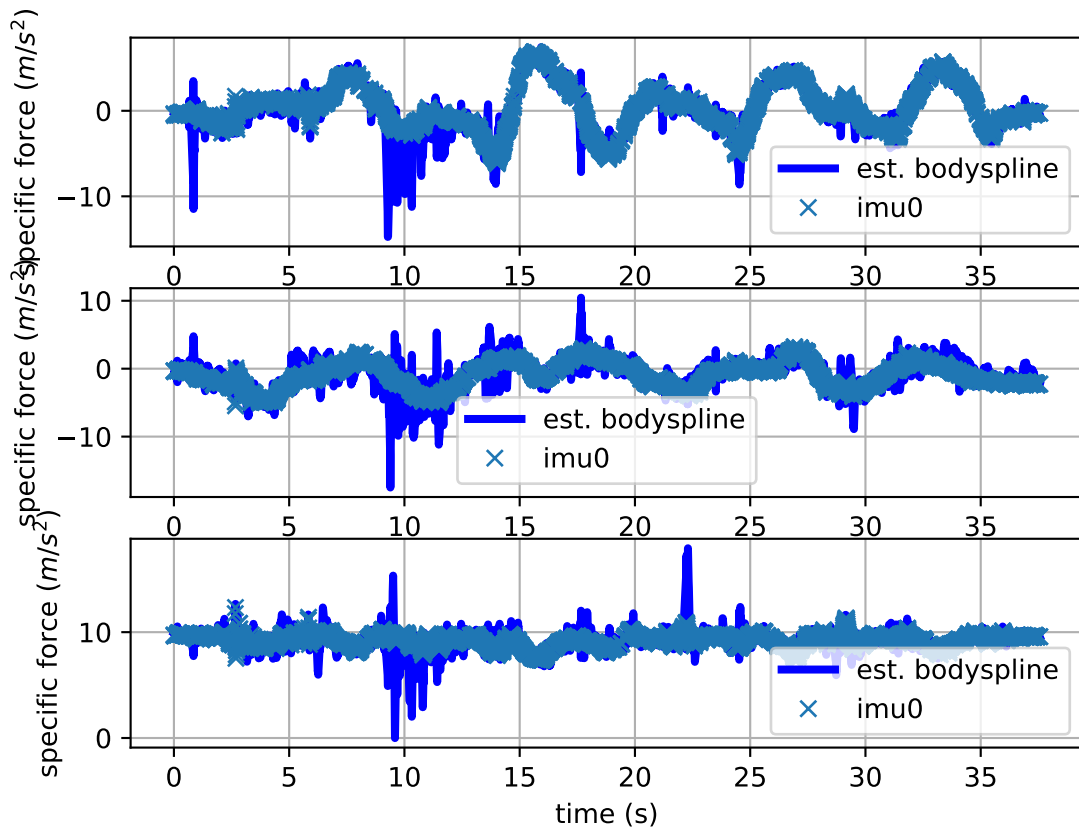
imu0: estimated poses



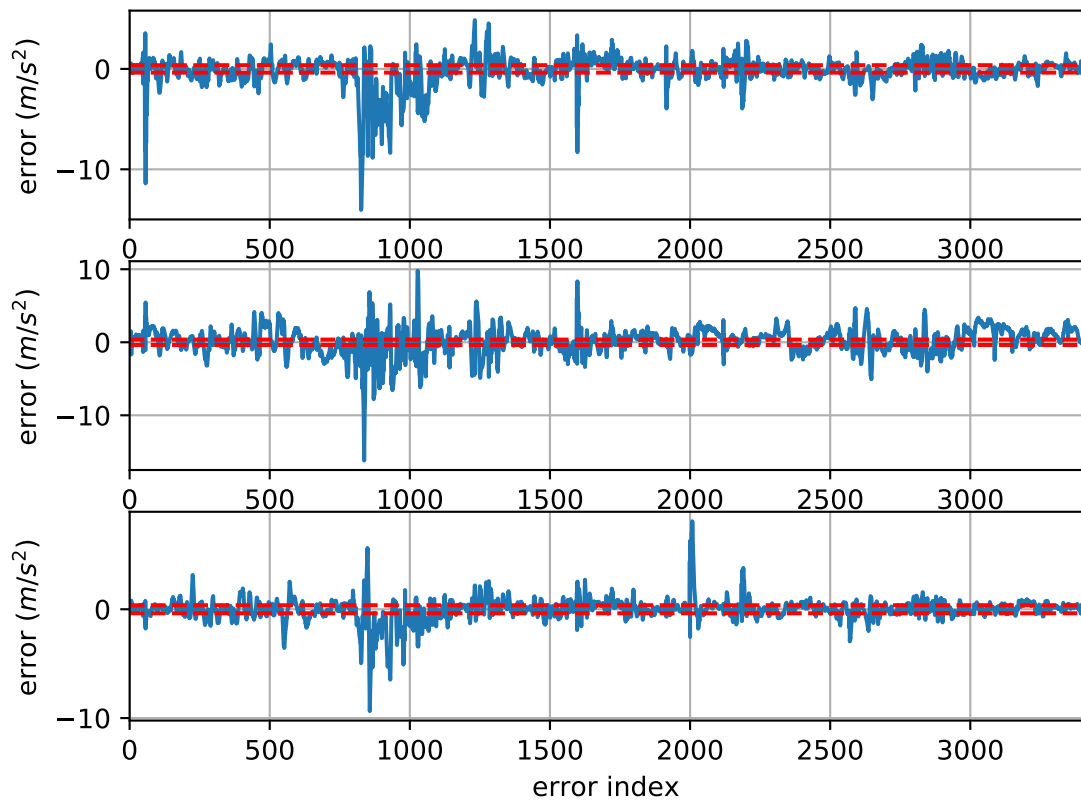
imu0: sample inertial rate



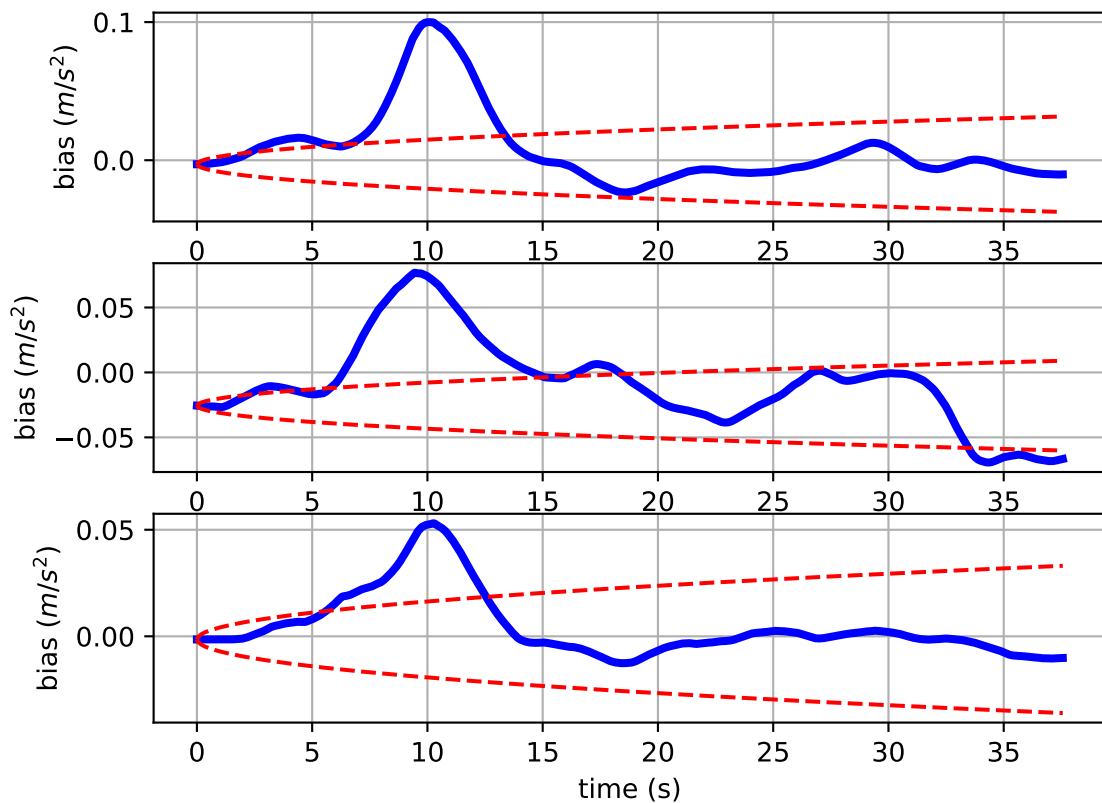
Comparison of predicted and measured specific force (imu0 frame)



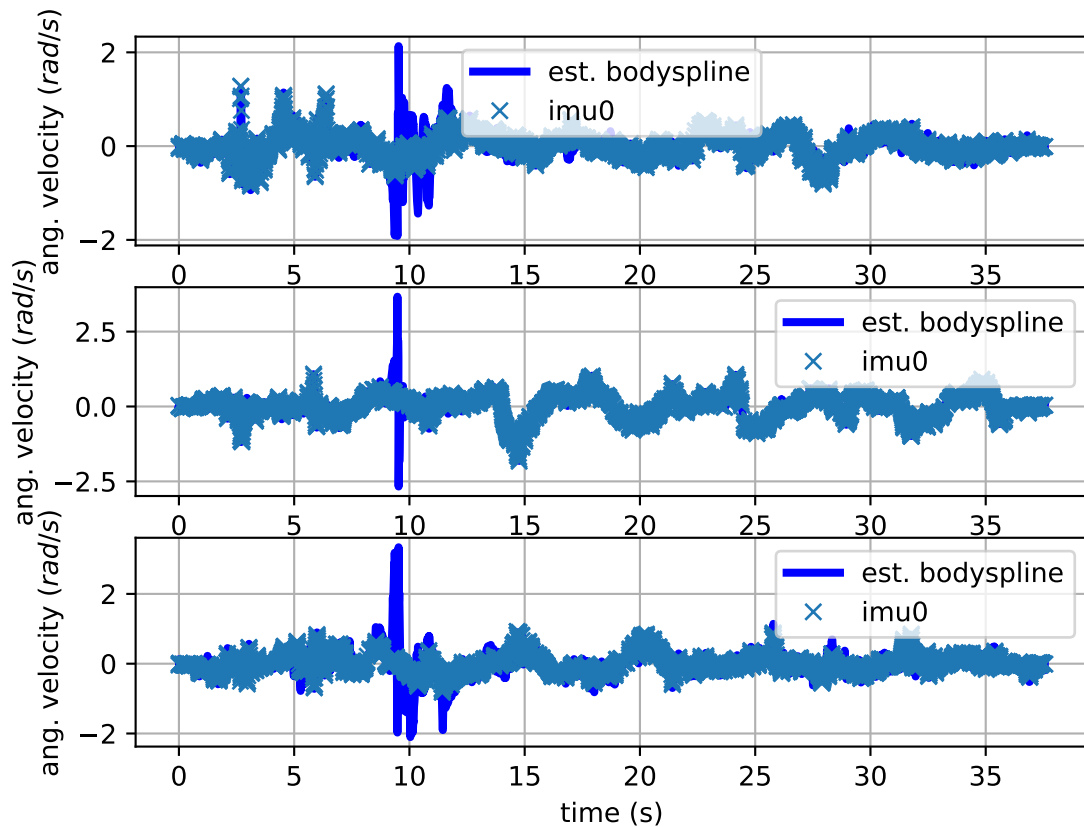
imu0: acceleration error



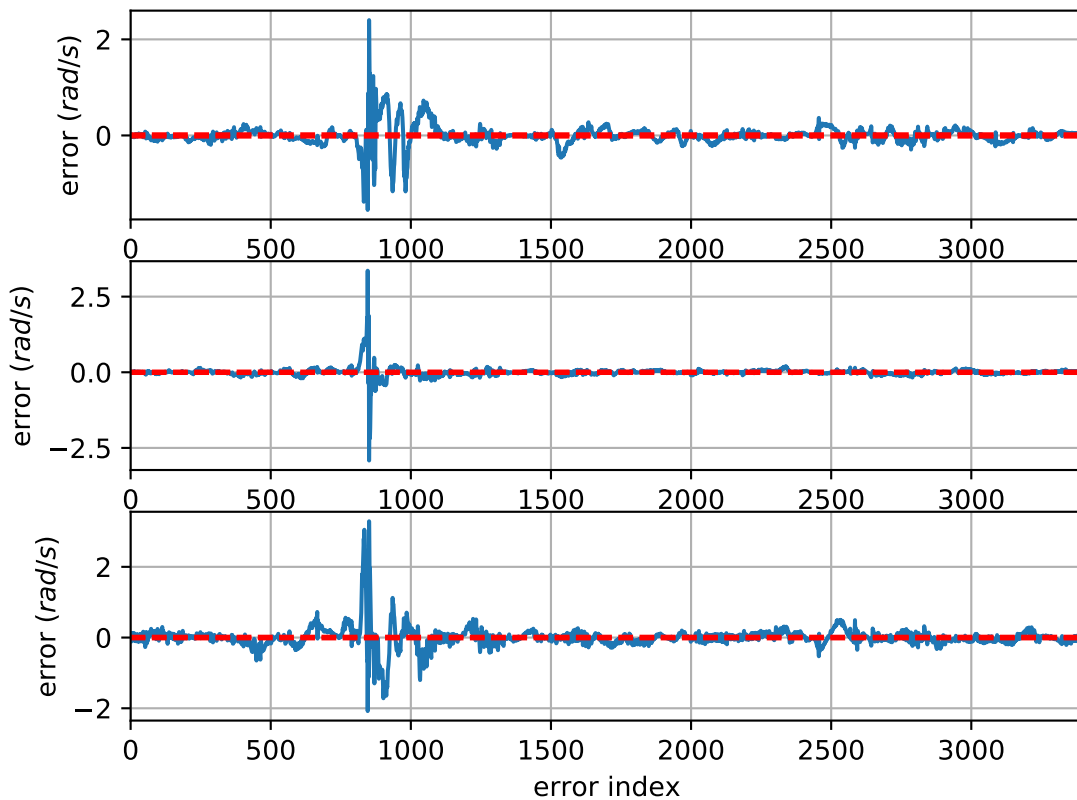
imu0: estimated accelerometer bias (imu frame)



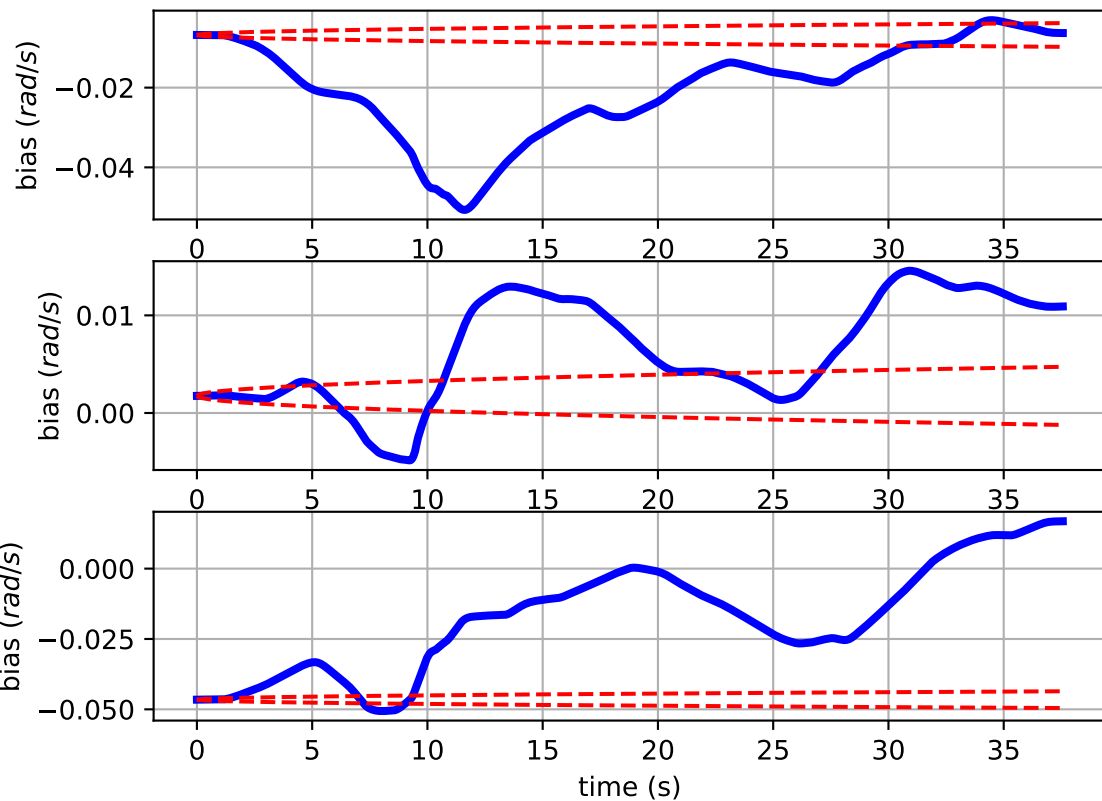
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

